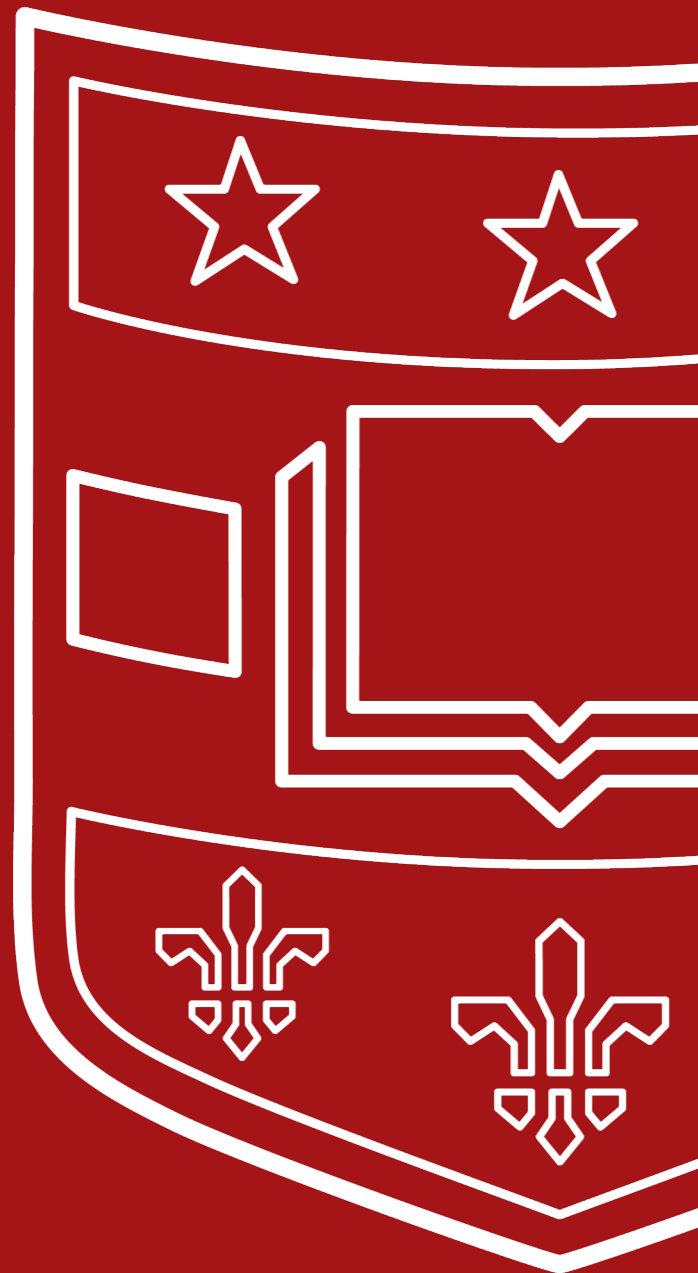


Approximate Private Inference in Quantized Models

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Zirui (Ken) Deng and Netanel Raviv

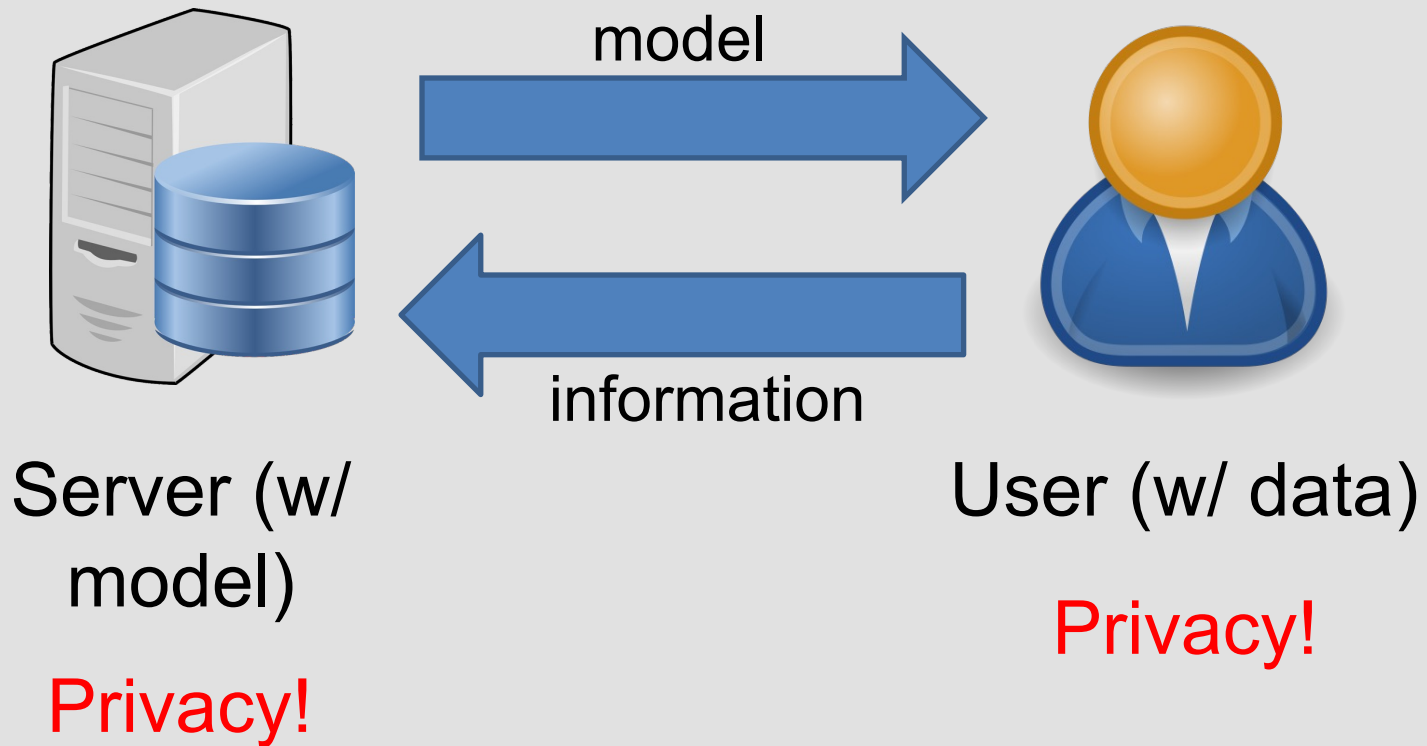


Motivation



- Loss of information privacy raises concerns.
 - Users rely on service providers for data management.
 - Trained models owned by service providers should also be protected.
- We focus on the *inference* stage in ML pipelines.

Private Inference





Previous Literature

- Homomorphic encryption (HE).
- ¹Raviv et al.: private inference as retrieval of inner products $\mathbf{w}\mathbf{x}^T$ between data and parameters.
- *Quantized* models: restricted parameter values.
 - $\mathbf{w} \in \{\pm 1\}^n$
 - Benefits: robustness² & efficiency³

¹N. Raviv, R. Bitar, and E. Yaakobi, “Information theoretic private inference in quantized models,” in 2022 IEEE International Symposium on Information Theory (ISIT), IEEE, 2022, pp. 1641–1646.

²A. Galloway, G. W. Taylor, and M. Moussa, “Attacking binarized neural networks,” in International Conference on Learning Representations, 2018.

³I. Hubara, M. Courbariaux, D. Soudry, R. El-Yaniv, and Y. Bengio, “Binarized neural networks: Training Neural Networks with Weights and Activations Constrained to +1 or −1,” in Proceedings of the 30th International Conference on Neural Information Processing Systems, 2016, pp. 4114–4122.



Previous Literature

- The server publishes a bit vector describing $\mathbf{w}$ while preserving privacy.
- The user computes several real vectors and their inner products with the data.
- The server combines answers to obtain $\mathbf{w}\mathbf{x}^T$.



Previous Literature

- Merit of such schemes: publication cost (number of bits published by the server) and privacy of server and user.
- Combined leakage from server and user has a lower bound, achieved by Raviv et al¹.
- In this paper we show that information leakage can be reduced upon **approximation** of inner products.

¹N. Raviv, R. Bitar, and E. Yaakobi, “Information theoretic private inference in quantized models,” in 2022 IEEE International Symposium on Information Theory (ISIT), IEEE, 2022, pp. 1641–1646.

Problem Statement



Server

Holds $\mathbf{w} \in \{\pm 1\}^n$
 $\mathbf{w} \sim W = \text{Unif}(\{\pm 1\}^n)$

Needs: $\mathbf{w}\mathbf{x}^T$

Privacy!

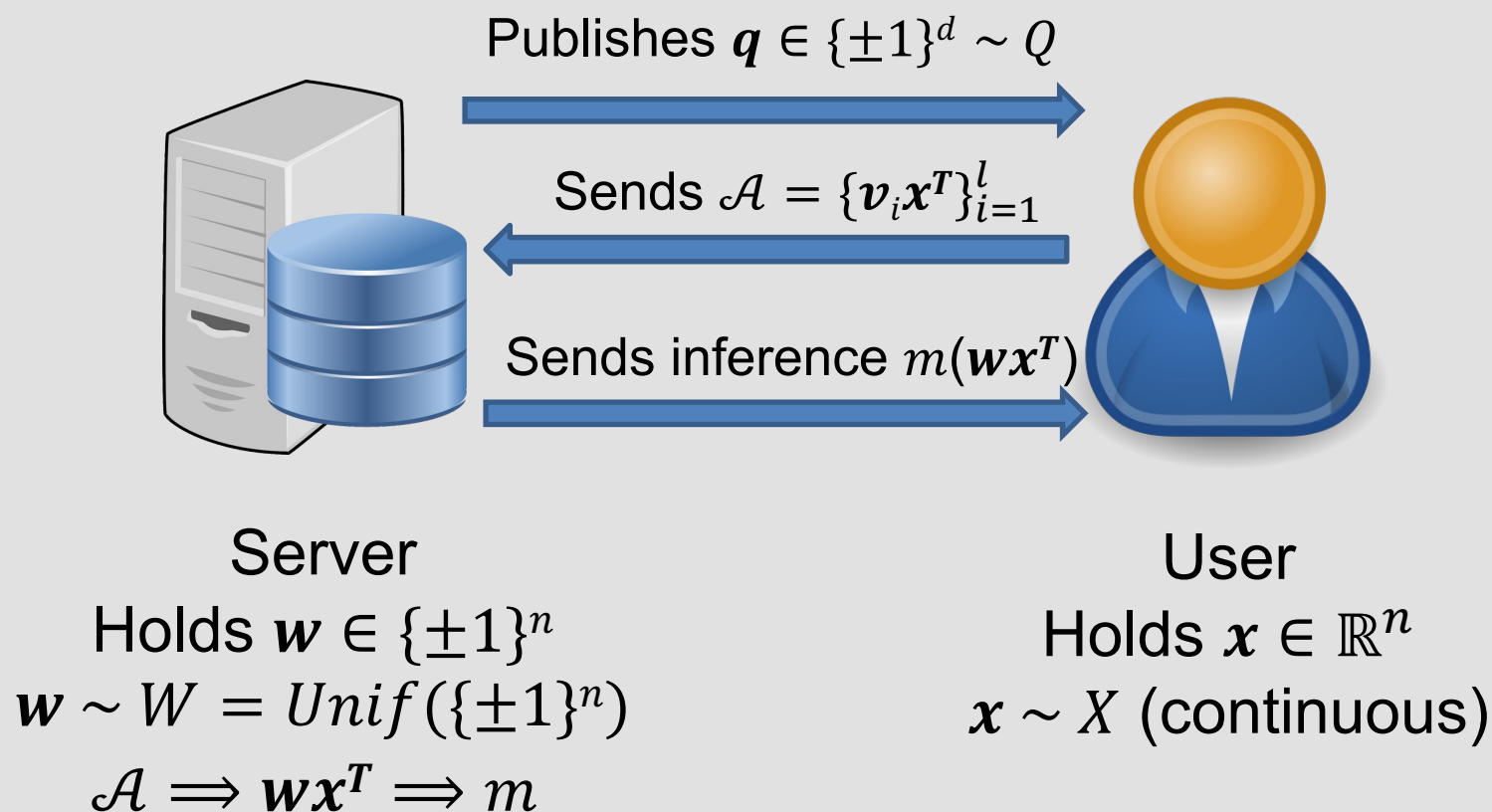


User

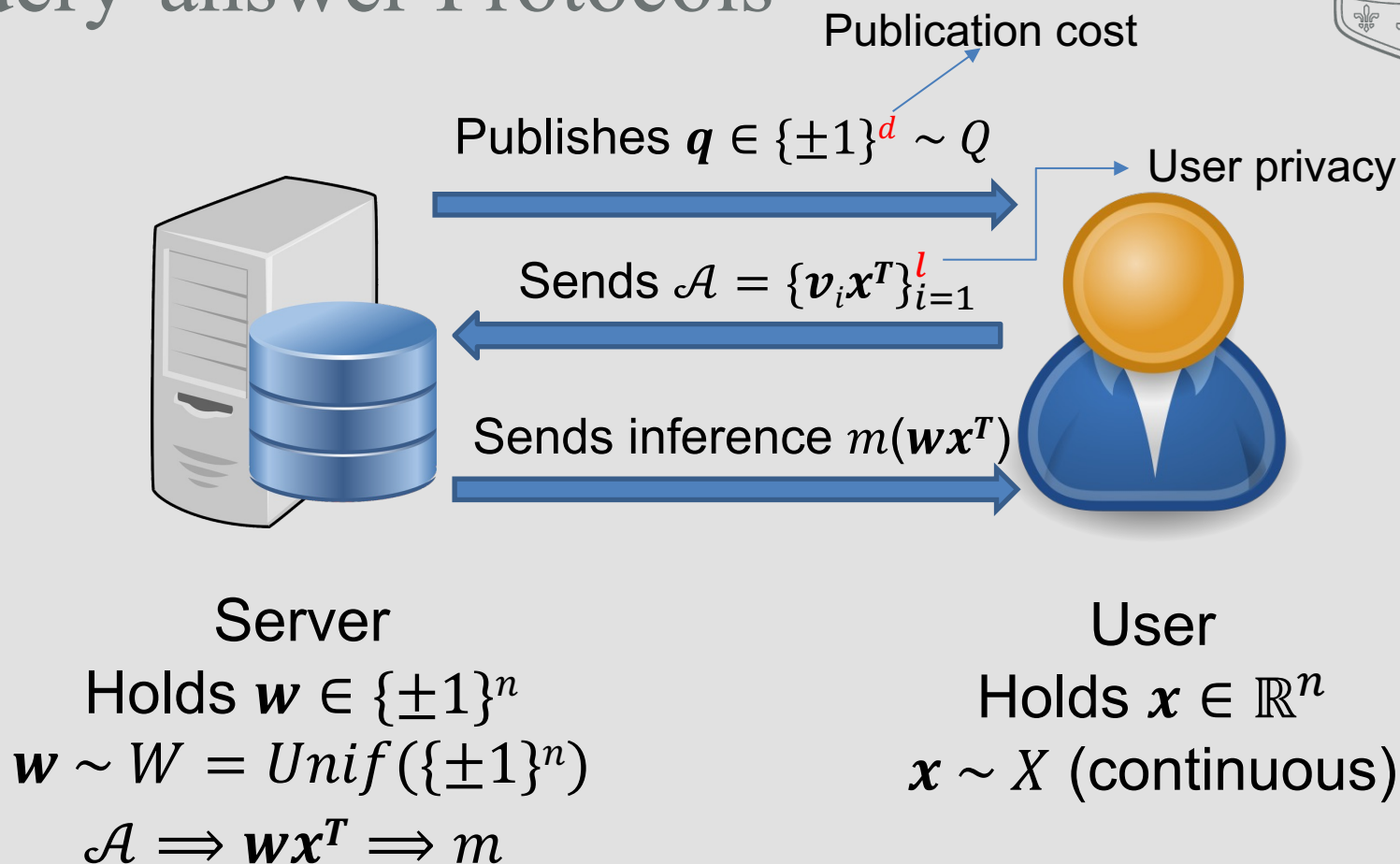
Holds $\mathbf{x} \in \mathbb{R}^n$
 $\mathbf{x} \sim X$ (continuous)

Privacy!

Query-answer Protocols



Query-answer Protocols



Server privacy: $I(W; Q) = H(W) - H(W|Q)$.



Previous Work

- Ideally, minimize $d, I(W; Q)$ and l simultaneously.
- However, ¹Theorem 1: $I(W; Q) + l \geq n$.
- t : privacy parameter ($1 \leq t \leq n$).
- ¹Publication cost $d = n - t$, user-privacy $l = t$, server-privacy $I(W; Q) = n - t$.
- $\Rightarrow$ Optimal scheme by Raviv et al.
- Requires *exact* computation of wx^T .

¹N. Raviv, R. Bitar, and E. Yaakobi, “Information theoretic private inference in quantized models,” in 2022 IEEE International Symposium on Information Theory (ISIT), IEEE, 2022, pp. 1641–1646.



Mathematical Background

- Let $\mathbb{F}_2$ be the binary field in $\{\pm 1\}$ representation. Let V be a subspace of $\mathbb{F}_2^n$.
- A **coset** of V is $V + \mathbf{a} \triangleq \{\mathbf{v} + \mathbf{a} \mid \mathbf{v} \in V\}$ for some $\mathbf{a} \in \mathbb{F}_2^n$, and $\mathbf{a}$ is called a *shift vector* that defines the coset.
- M : full-rank matrix defined over $\mathbb{F}_2$ such that $V = \ker M$.
- A **syndrome** of a vector $\mathbf{u} \in \mathbb{F}_2^n$ is the vector $M\mathbf{u}^T \in \mathbb{F}_2^{n-\dim V}$.
- Vectors have the same syndrome $\Leftrightarrow$ they belong to the same coset.

Overview of Approximation Scheme



Privacy parameter $t \in [n]$.

$S = \{S_1, \dots, S_t\}$, a partition of $[n]$.

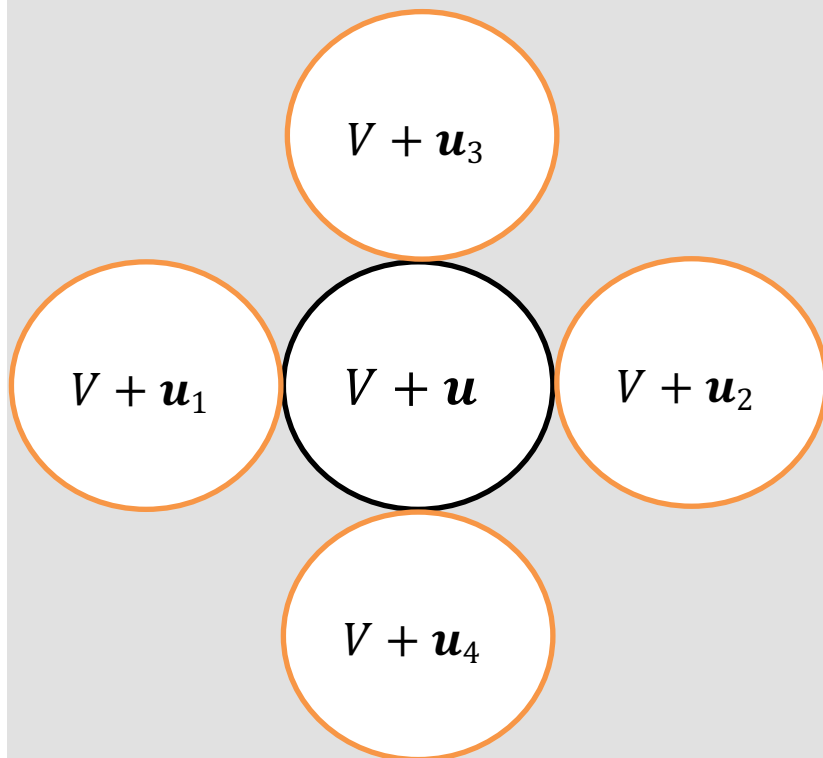
$\mathbb{1}_{S_i}$ s.t. $(\mathbb{1}_{S_i})_j = -1$ if $j \in S_i$ and 1 otherwise.

$V = \text{Span}_{\mathbb{F}_2} \{\mathbb{1}_{S_i}\}_{i=1}^t$.

Overview of Approximation Scheme



- Previously, extract signal $\mathbf{w}\mathbf{x}^T$ with $\mathbf{w} \in V + \mathbf{u}$.
- Now, retrieve approximate signal $\mathbf{w}'\mathbf{x}^T$, where $\mathbf{w}' \in V + \mathbf{u}'$ with $V + \mathbf{u}' \neq V + \mathbf{u}$ but close to $V + \mathbf{u}$.



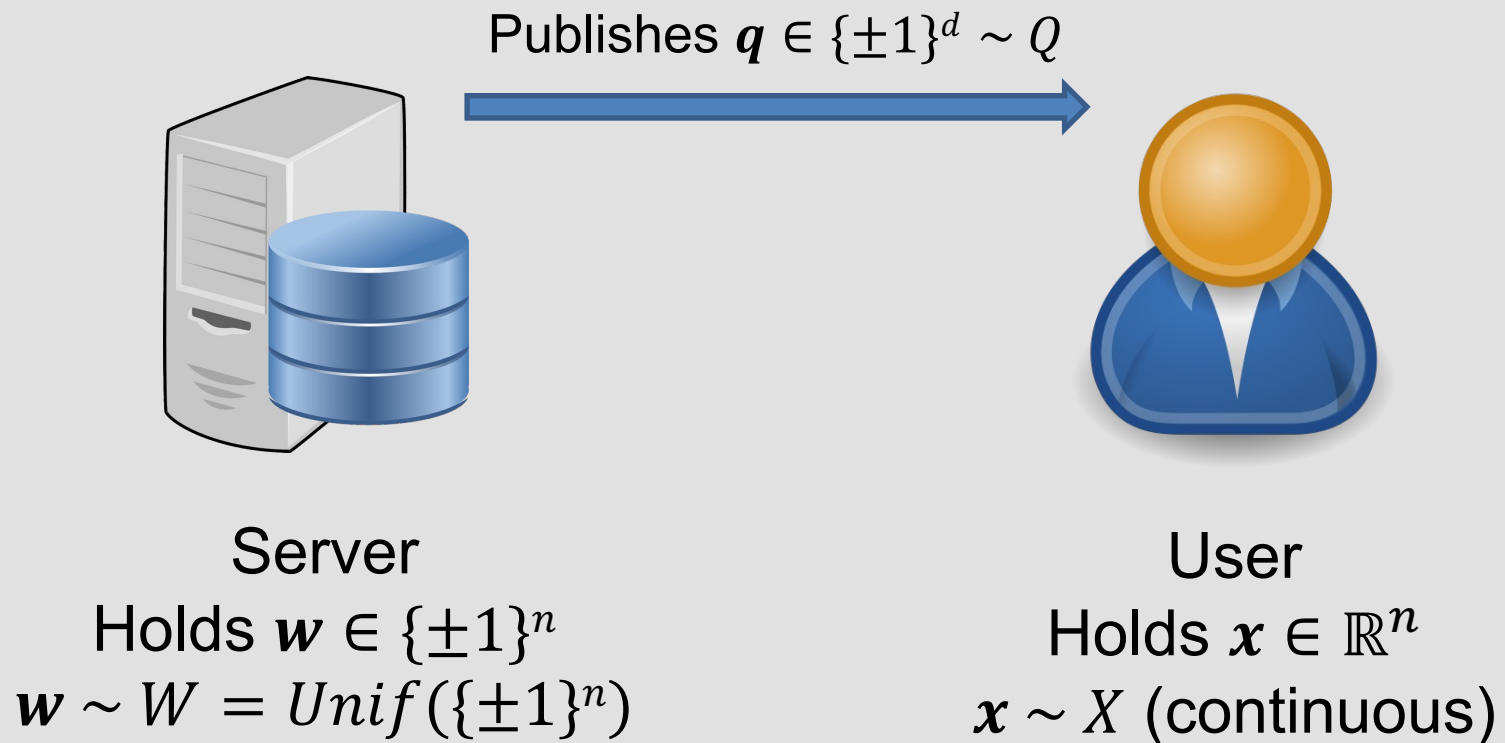
- Randomly choose adjacent coset
- More uncertainty to $W|Q$
- Higher $H(W|Q)$
- Lower $I(W; Q) = H(W) - H(W|Q)$.

Overview of Approximation Scheme



- Quality of approximation hinges on $\frac{|\mathbf{w}'x^T - \mathbf{w}x^T|}{2\|\mathbf{w}\|_2\|\mathbf{x}\|_2}$, the normalized difference between approximate and true signal.
- Lemma 1: If $\|\mathbf{x}\|_2 \leq c$, then $|\mathbf{w}'x^T - \mathbf{w}x^T| \leq c \cdot \|\mathbf{w} - \mathbf{w}'\|_2$ for every $\mathbf{w}'$.
- Goal: Find $\mathbf{w}'$ closest to $\mathbf{w}$ in Euclidean distance.

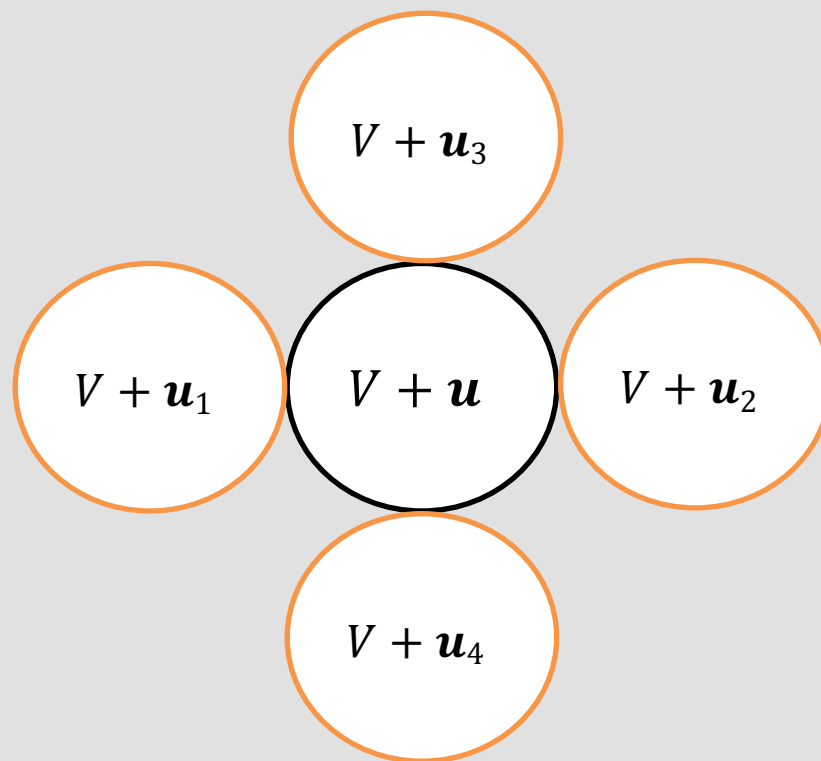
The Approximation Algorithm



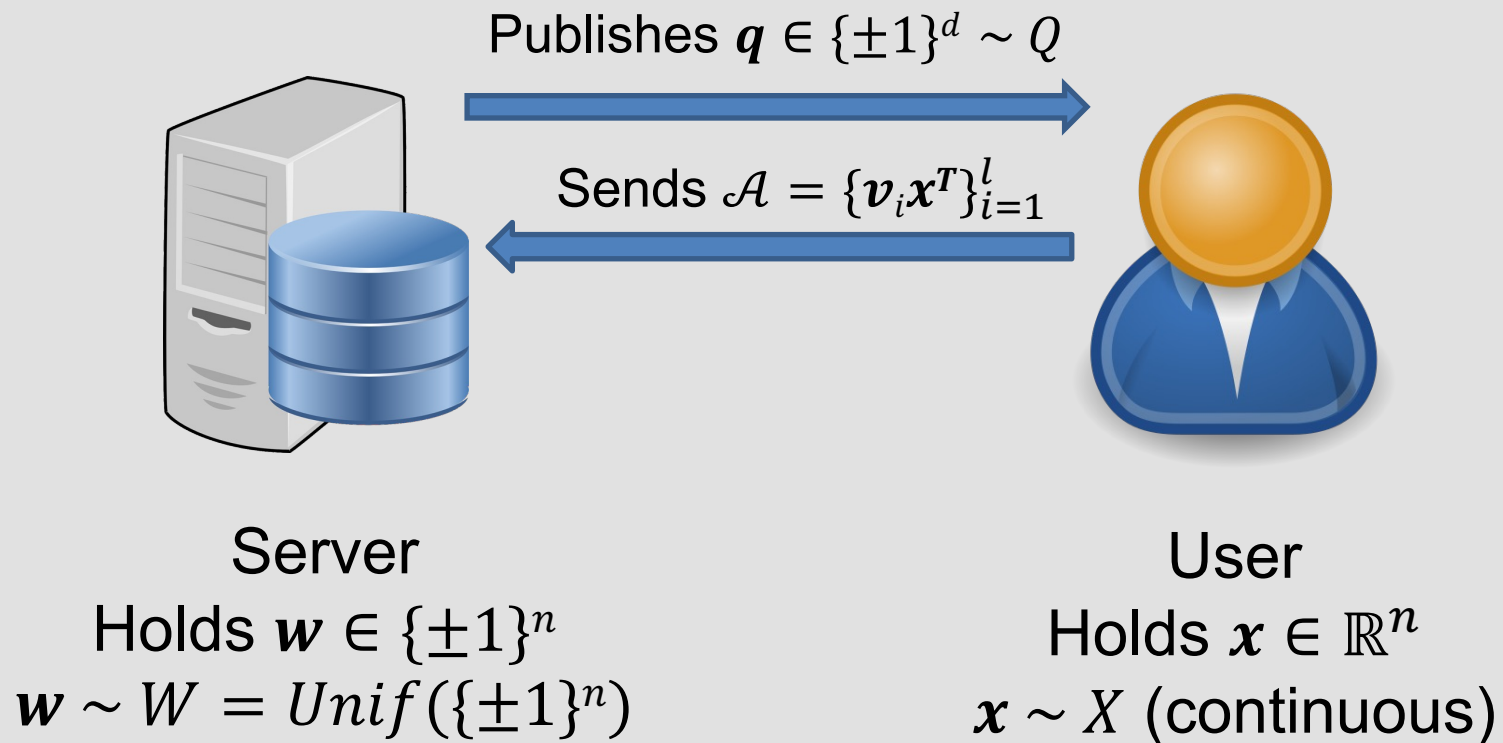


The Approximation Algorithm

- The server chooses $\mathbf{u}_i$ uniformly at random and publishes $\mathbf{q}^T \triangleq M(\mathbf{w} + \mathbf{u}_i)^T \in \mathbb{F}_2^{n-t}$.



The Approximation Algorithm

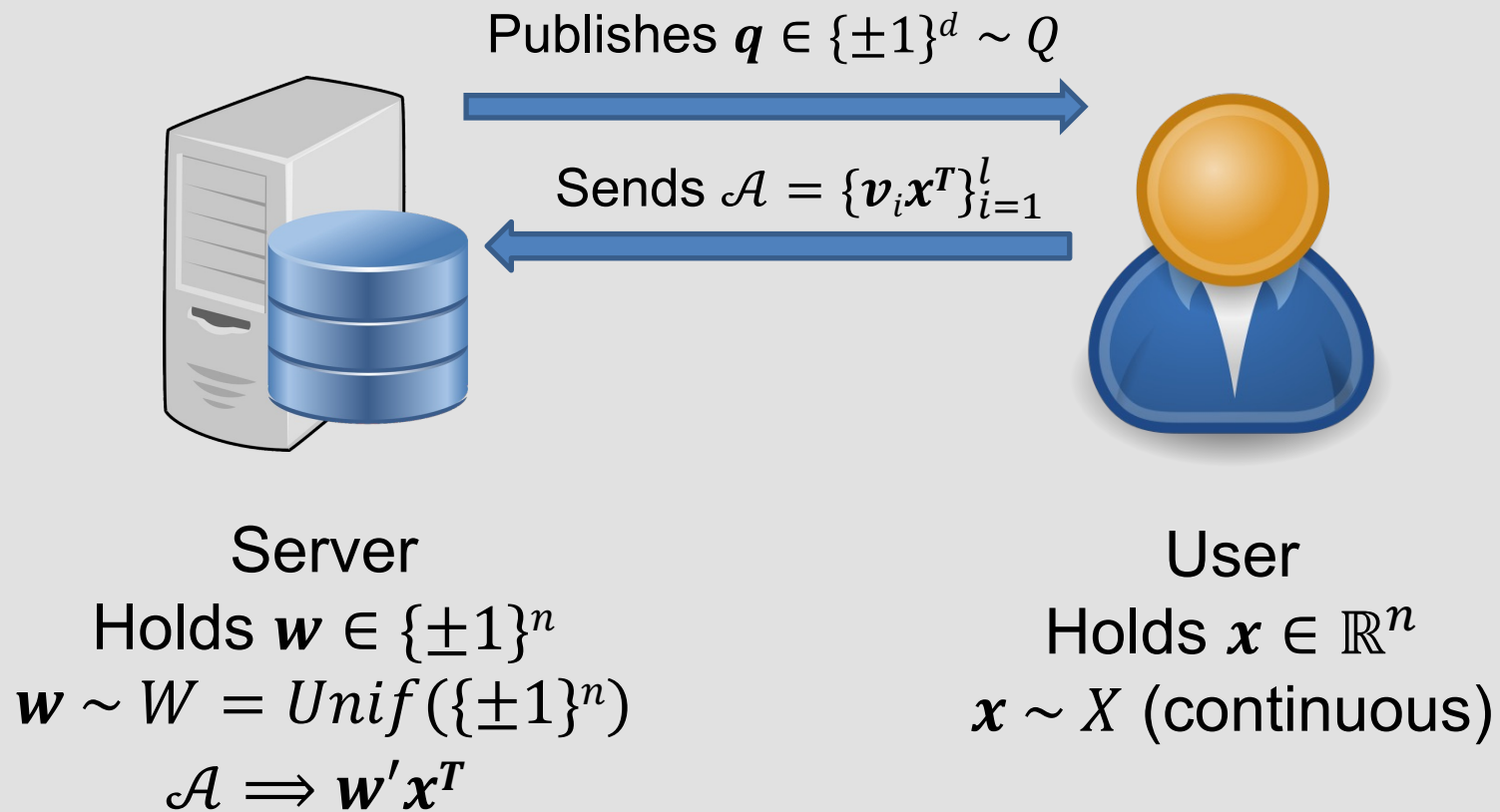




The Approximation Algorithm

- The user defines $\mathbf{u}' \triangleq B(M, \mathbf{q})$,
- defines $\mathbf{v}_i$ based on $\mathbf{u}'$ and the partition $S = \{S_1, \dots, S_t\}$ for every $i \in [t]$,
- and then sends $\mathcal{A} = \{\mathbf{v}_i \mathbf{x}^T\}_{i=1}^t$ to the server.

The Approximation Algorithm





The Approximation Algorithm

- The server also computes $\mathbf{u}' \triangleq B(M, \mathbf{q})$,
- computes the appropriate $\mathbf{w}'$ closest to $\mathbf{w}$ based on $\mathbf{u}'$,
- and outputs $\mathbf{w}'\mathbf{x}^T$.



Merit of Approximation Algorithm

- Publication cost $d = n - t$.
- User-privacy $l = t$.
- Server-privacy is improved at the cost of approximate computation of $\mathbf{w}\mathbf{x}^T$.
- Theorem 2: good approximation from protocol.
- Theorem 3: improved privacy of the server.



Merit of Approximation Algorithm

- Theorem 2: In our approximation scheme,

$$\frac{|w'x^T - wx^T|}{2||w||_2||x||_2} \leq \frac{\sqrt{h(1-h/n)}}{\sqrt{n}}, \text{ where } h \geq w_H(\mathbf{g}) \text{ for any shift vector } \mathbf{g}.$$

- RHS vanishes with $n \Leftrightarrow h = o(n)$.
- Difference (error) between approximate and true signal is relatively negligible.
- Our algorithm gives good approximation.

Merit of Approximation Algorithm



- Theorem 3: In our approximation scheme, $I(W; Q) = n - t - \log(\# \text{ adjacent cosets})$.
 - $I(W; Q) = n - t$ in the exact scheme.
- Case 1: Vanishing error between approximate and true signal $\rightarrow$ sublinear improvement in $I(W; Q)$.
- Case 2: Non-vanishing error $\rightarrow \Omega(n)$ improvement.

Summary



- We view private inference as a task of retrieving signals from data.
- We relax computation of inner products and provide better privacy guarantees (vanishing/non-vanishing error) while ensuring good approximation of signals.
- Future directions include deriving a comprehensive lower bound for server privacy.

Thank you!

